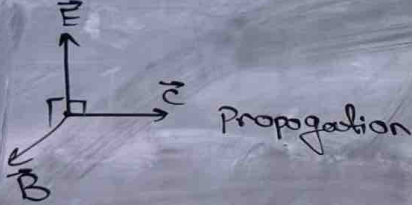


بسم الله الرحمن الرحيم

Electromagnetic Wave

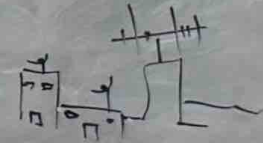
EM wave consists of E and B perpendicular to each other and perpendicular to the propagation direction.



$v \rightarrow$ velocity through a medium
 $c \rightarrow$ velocity in free space

characteristics of EM waves:

- ① EM waves generated by oscillating charges
- ② EM wave energy depends on a frequency and wavelength



- ③ All EM waves propagate with a speed of light $c = 3 \times 10^8 \text{ m/s}$
 $E = hf$, $E = \frac{hc}{\lambda}$
- ④ All EM waves are chargeless
 $\rightarrow$ not deflected in E and B